

IOWA



YouTube Comment Sentiment Analysis

By: Nick Hageman & Nate Schaefer

April 7th, 2025

-  **@rudedog6868** 7 days ago
I'm an Apple guy. But you can't deny Apple's arrogance and myopia kicked them in the ass. Neg
1.1K  [Reply](#)
20 replies
-  **@LaurenGlenn** 9 days ago
One of the coolest things that Cortana used to do on Windows Phone was something like "Cort Walmart". The second we walked in the door of Walmart, it said, "Buy Toilet Paper" on the screen.
That was the kind of AI that is actually useful..... I don't think any other app has done something like that.
Read more
16K   [Reply](#)
328 replies
-  **@ArthurMoore** ✓ 9 days ago
Hey Siri, when is apple intelligence coming out?
Siri: "Sorry, I don't understand"
11K  [Reply](#)
58 replies
-  **@Chef7Curry** 3 days ago
The race is on, GTA VI vs Apple Intelligence 🤖
143  [Reply](#)
1 reply
-  **@bjk837** 8 days ago
Steve Jobs warned about what could happen to a company when the marketing department starts talking about AI.
4.5K  [Reply](#)
47 replies
-  **@kasratabrizi2839** 7 days ago
Man I recently used Siri and Apple Intelligence to do something extremely easy: Change the title of a video.
This is how it went:
-
Read more
747  [Reply](#)
13 replies
-  **@Green_Lantern89** 9 days ago
Apple bought the Siri app YEARS ago and still have the worst AI assistant. Inexcusable.
8.2K  [Reply](#)
120 replies
-  **@Louise-fix** 6 days ago
Siri's response is still "here's what I found on the web" instead of simply telling you the answer, like Google Assistant.
887  [Reply](#)
9 replies
-  **@MasukSarkerBatista** ✓ 9 days ago
Never buy a product based on future promises

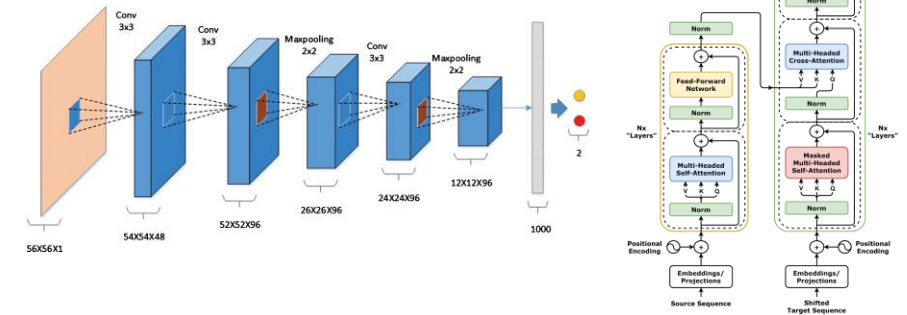
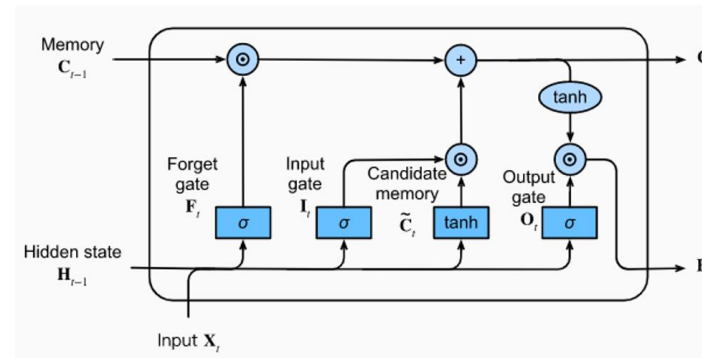
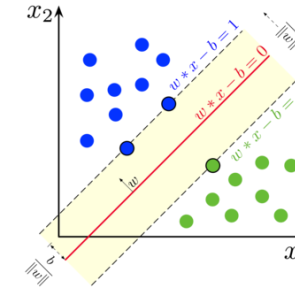
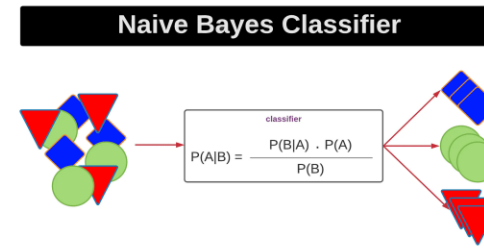
Introduction

- Text classification to classify YouTube comments as positive, negative, or neutral
- Benefits user engagement and detecting harmful content
- Challenges include an imbalanced dataset, emojis, slang, special characters
- Provide content creators with insights into viewer reactions



Literature Review

- Earlier studies explored use of:
 - Naive Bayes
 - Support Vector Machines
 - Random Forest
- More recent studies have used:
 - LSTMs
 - CNNs
 - Transformers



Methodology

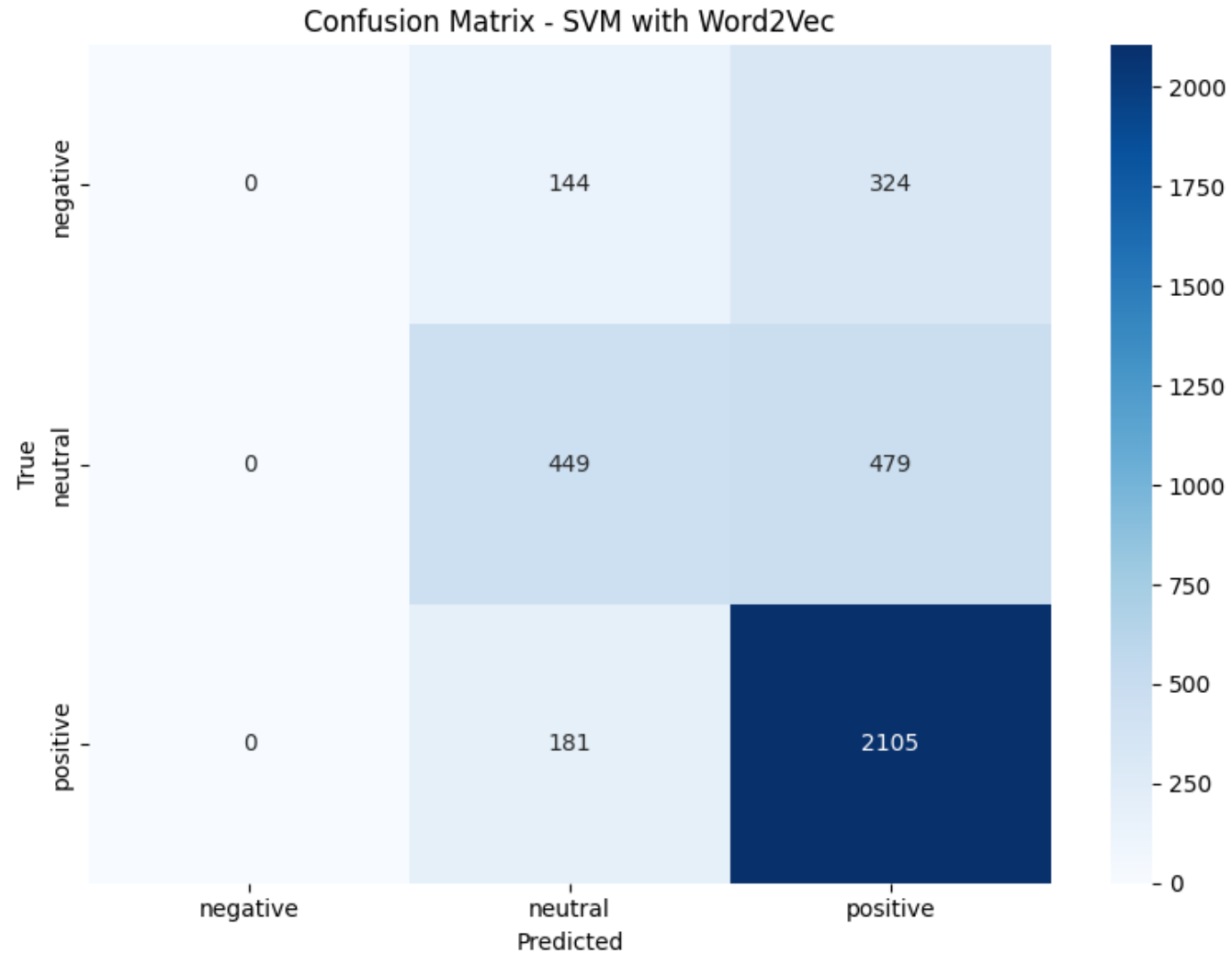
- Planned to use traditional machine learning methods
 - NB, SVM, and RF
- Also include advanced methods
 - SVM with Word2Vec
 - SVM with Co-occurrence SVD
- Performance evaluators
 - Accuracy, precision, recall, f1 score

Experiments

- Performance metric table

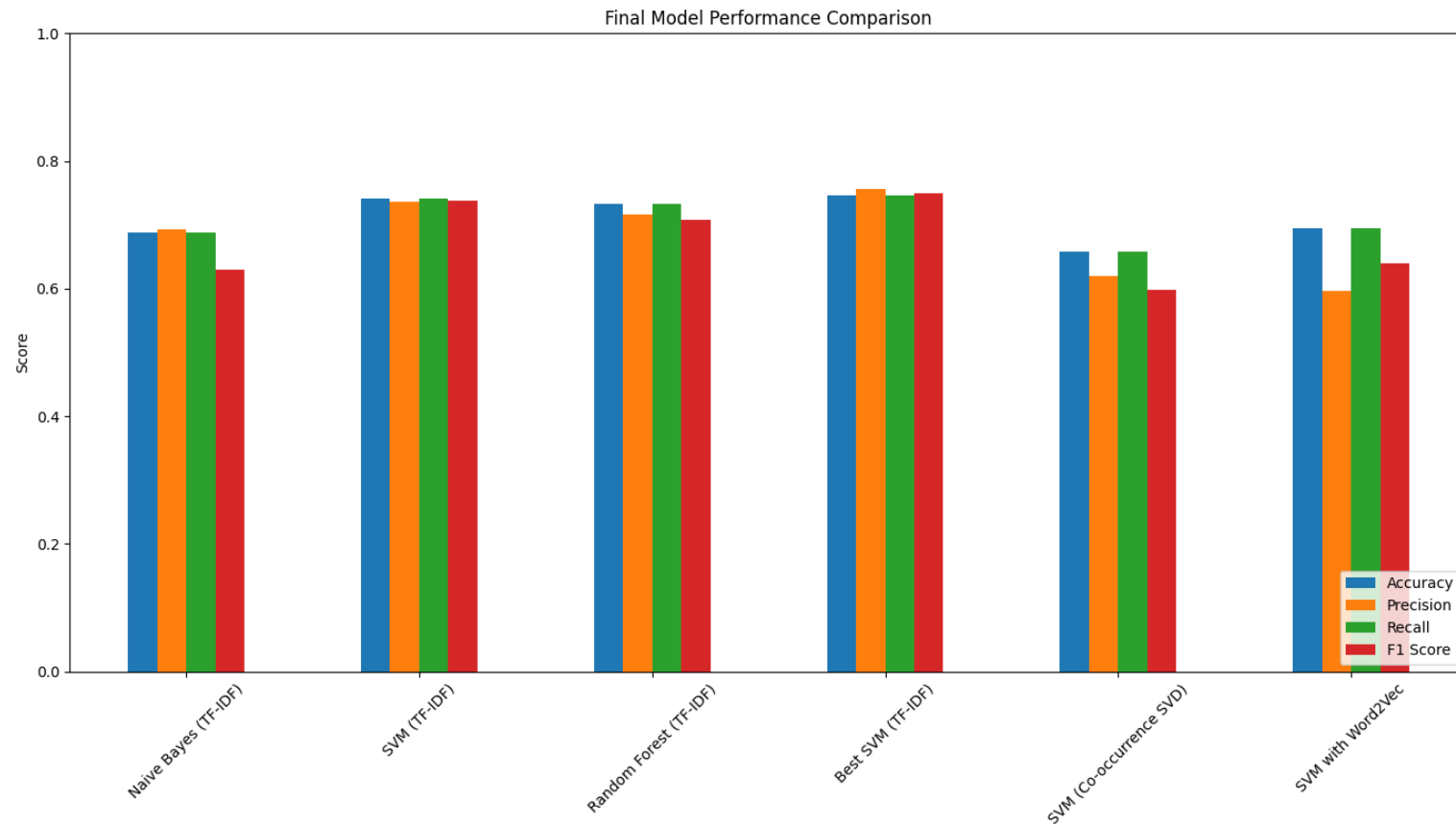
Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes (TF-IDF)	0.6874	0.6927	0.6874	0.6303
SVM (TF-IDF)	0.7412	0.7359	0.7412	0.7379
SVM (Word2Vec)	0.6942	0.5963	0.6942	0.6376
SVM (Co-occurrence SVD)	0.6583	0.6204	0.6583	0.5975
Best SVM (TF-IDF)	0.7452	0.7567	0.7452	0.7501
RF (IF-IDF)	0.7333	0.7166	0.7333	0.7070

Experiments

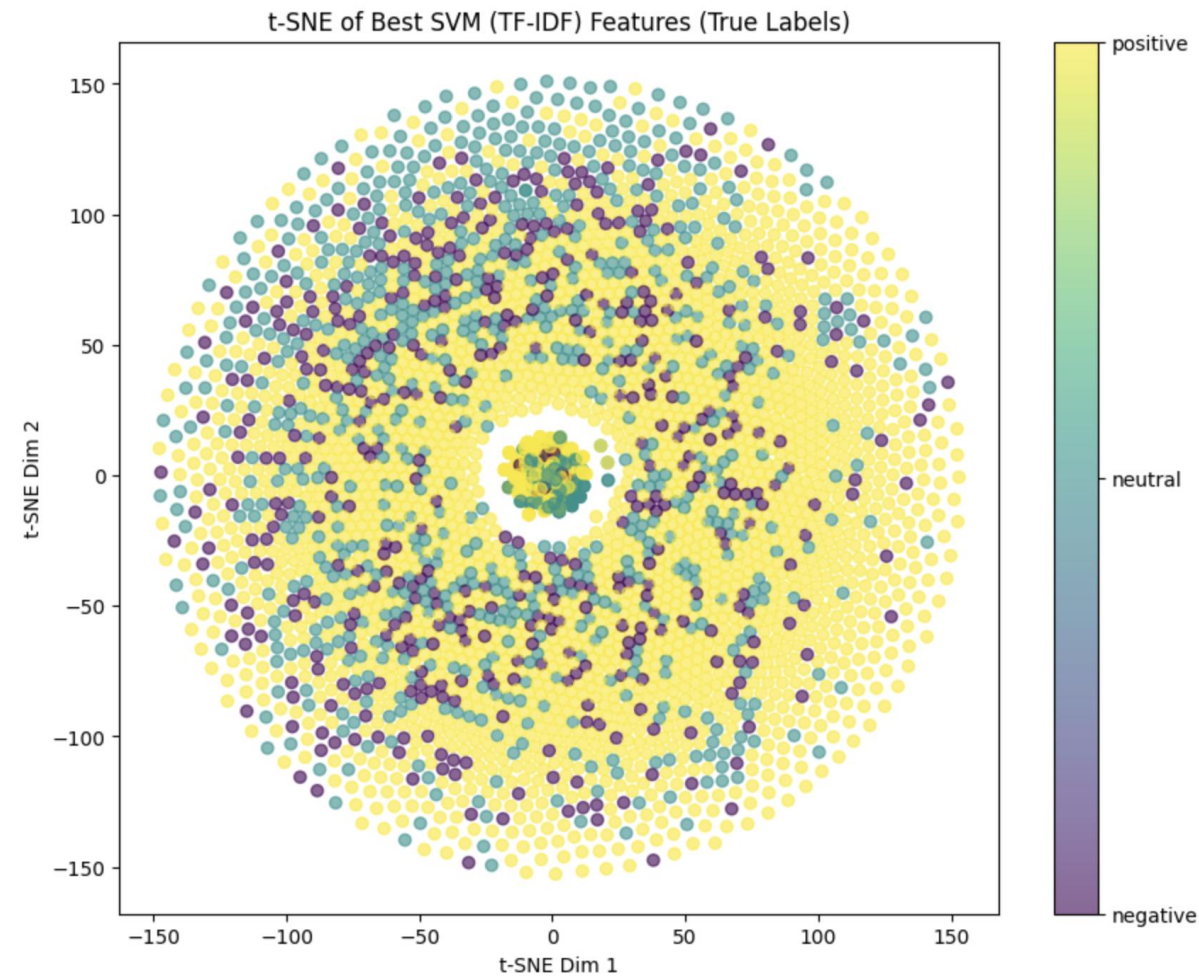
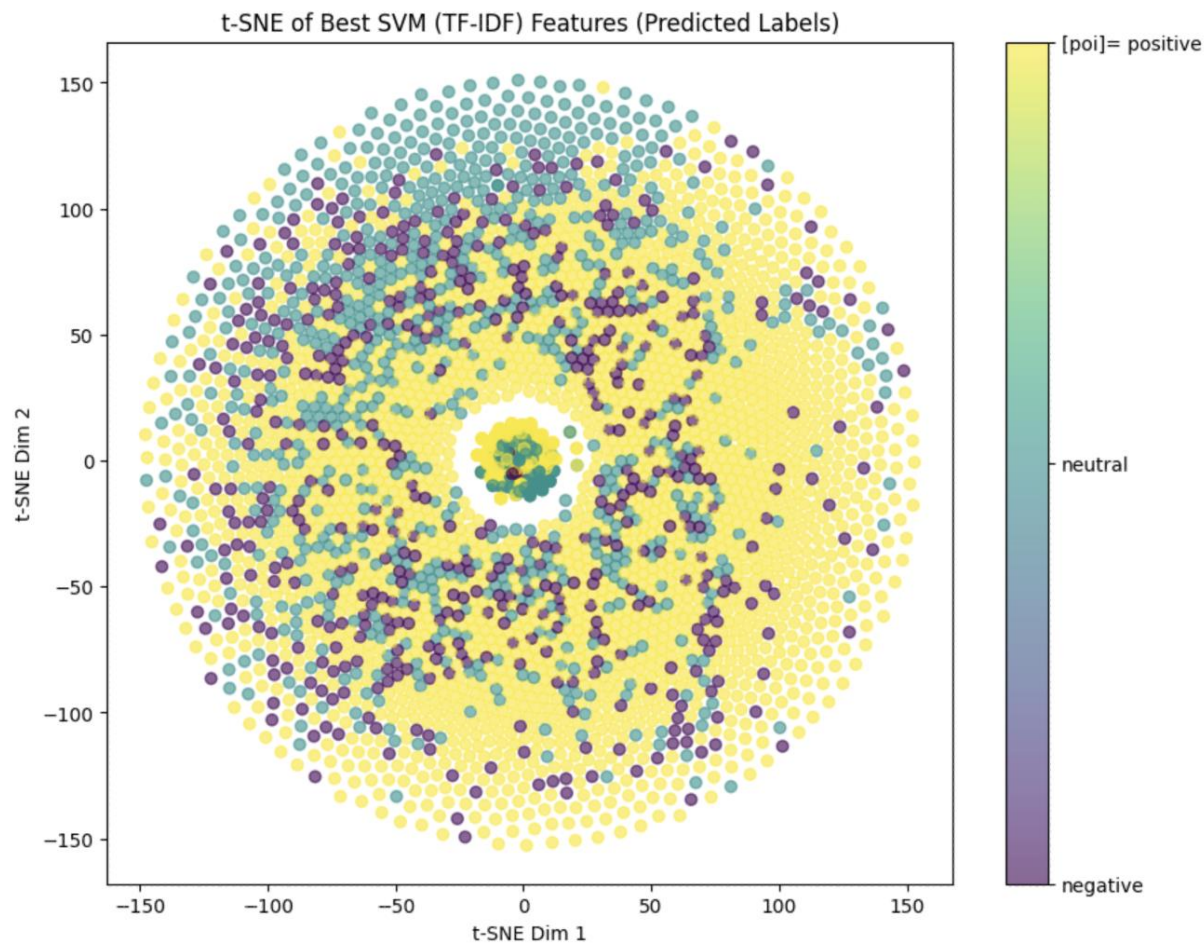


Experiments

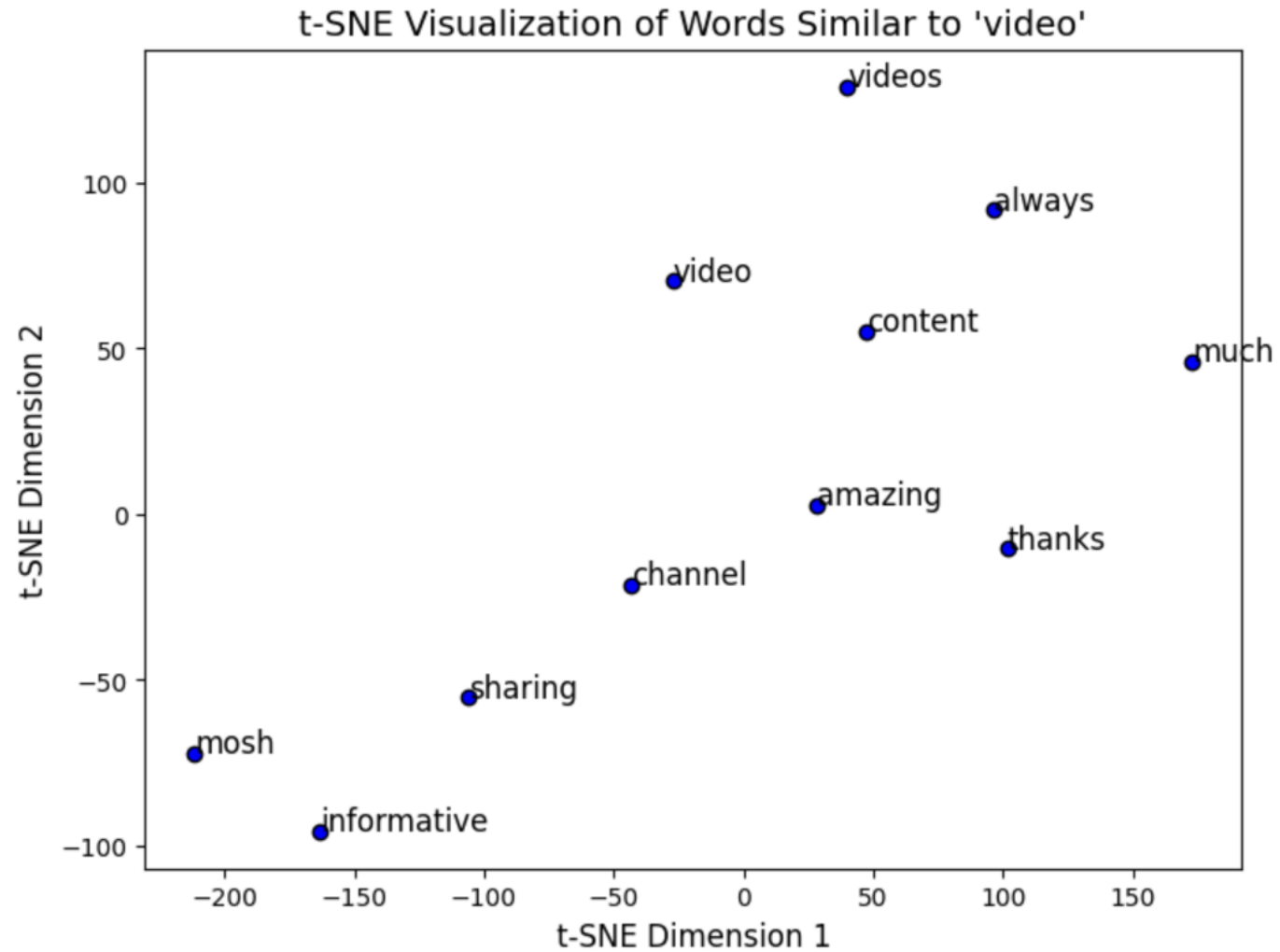
- Performance metric graph



Experiments



Experiments



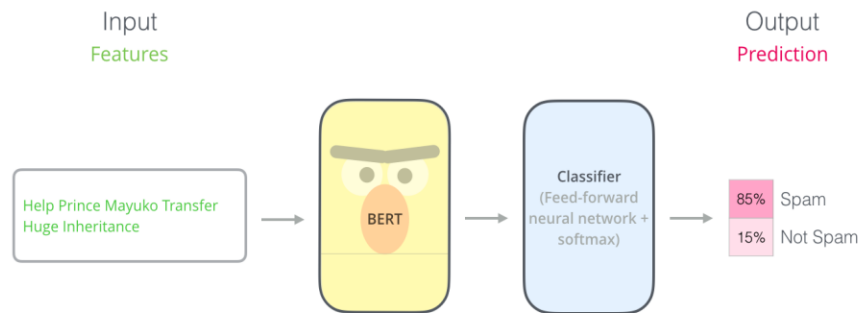
Experiments

- Misclassifications
 - We observed that foreign languages are often counted positive
 - Even humans manually annotating these would have a hard time

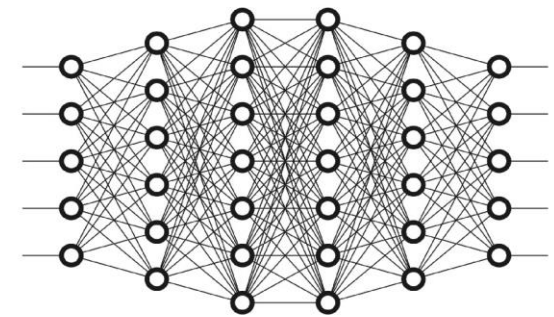
Comment	True Sentiment	Predicted Sentiment
Imagine scammer get hurt someone points win	negative	positive
Bihar ba	neutral	positive
Watching asmr ramadan	neutral	positive
Didn't believe anything wrong refusing turn doctor	negative	neutral
bài của buffalove thực sự quá hay đi ạ em	neutral	positive
Since little knew I'd rather live street become	negative	neutral
You're friends zhc interesting	neutral	positive

Conclusion

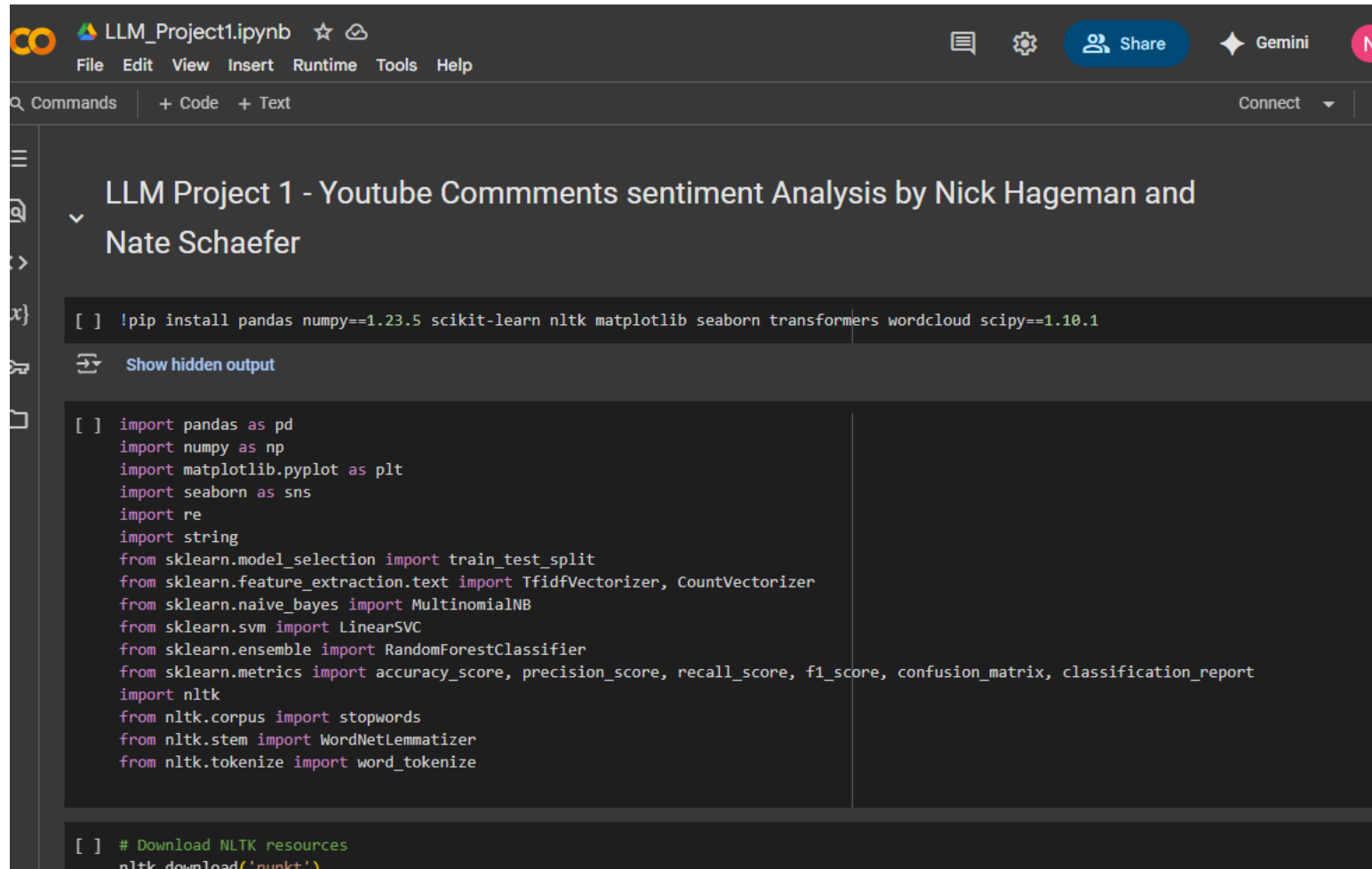
- SVM model with hyperparameter tuning performed the best in all performance metrics
- Future work could include incorporating more pre-trained models such as BERT



- Also could experiment with other deep learning models too



Code Overview



The screenshot displays a Jupyter Notebook titled "LLM_Project1.ipynb". The interface includes a top menu bar with options like File, Edit, View, Insert, Runtime, Tools, and Help. Below the menu, there's a toolbar with icons for commands, code, and text, along with a "Connect" button. The notebook content is titled "LLM Project 1 - Youtube Comments sentiment Analysis by Nick Hageman and Nate Schaefer". It contains two code cells. The first cell installs various Python libraries: pandas, numpy, scikit-learn, nltk, matplotlib, seaborn, transformers, wordcloud, and scipy. The second cell imports these libraries and sets up the environment for sentiment analysis, including imports for train_test_split, TfidfVectorizer, CountVectorizer, MultinomialNB, LinearSVC, RandomForestClassifier, accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report, and nltk resources.

```
[ ] !pip install pandas numpy==1.23.5 scikit-learn nltk matplotlib seaborn transformers wordcloud scipy==1.10.1
```

Show hidden output

```
[ ] import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import re
import string
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer, CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import LinearSVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize
```

```
[ ] # Download NLTK resources
nltk.download('punkt')
```

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